

Project Design Phase-II Technology Stack (Architecture & Stack)

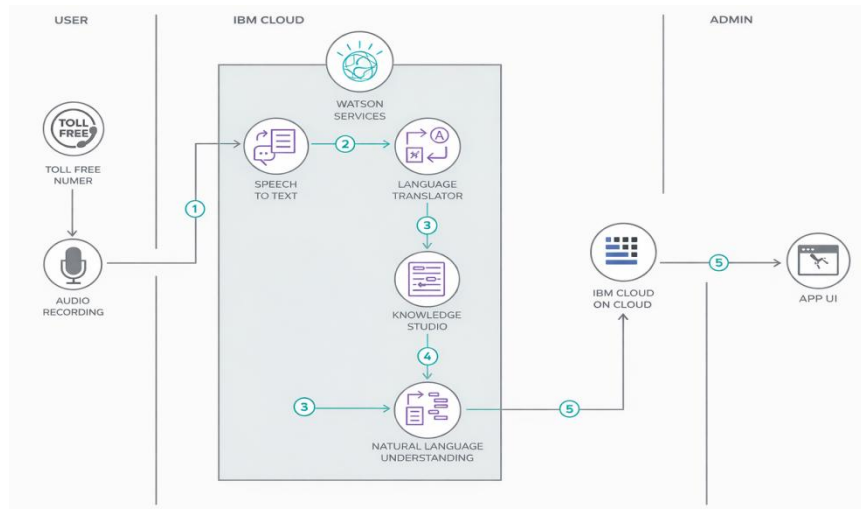
Date	19 February 2026
Team ID	LTVIP2026TMIDS62122
Project Name	Startup Success Prediction System Using Machine Learning
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the application (Web UI, Mobile App, Chatbot, etc.)	HTML, CSS, JavaScript, Angular JS, React JS
2	Application Logic – 1	Core business logic handling application processes	Java / Python
3	Application Logic – 2	Speech processing logic	IBM Watson Speech to Text (STT) Service
4	Application Logic – 3	Conversational AI / Chatbot logic	IBM Watson Assistant
5	Database	Stores structured data, configurations, and application records	MySQL, NoSQL
6	Cloud Database	Managed cloud-based database service	IBM DB2, IBM Cloudant
7	File Storage	Storage for uploaded files, documents, audio files, etc.	IBM Block Storage / Other Cloud Storage / Local Filesystem
8	External API – 1	Integration with third-party service	IBM Weather API
9	External API – 2	Government / Identity verification API integration	Aadhaar API
10	Machine Learning Model	AI/ML model used in the system	Object Recognition Model
11	Infrastructure (Server / Cloud)	Deployment environment of the application	Local Server / Cloud Foundry / Kubernetes

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List of open-source frameworks used in the application for development and deployment.	React JS / Angular JS (Frontend), Node.js / Python (Backend), Express.js, Spring Boot
2	Security Implementations	Security mechanisms implemented including authentication, authorization, encryption, firewall protection, and vulnerability prevention.	SHA-256 Password Hashing, AES Encryption, HTTPS (SSL/TLS), JWT Authentication, OAuth 2.0, IAM Controls, OWASP Security Standards, Firewall
3	Scalable Architecture	Architecture designed to handle increasing users and workload efficiently (e.g., 3-Tier or Microservices architecture).	Microservices Architecture, Docker Containers, Kubernetes, IBM Cloud Foundry
4	Availability	Ensures system uptime and fault tolerance using redundancy and load balancing.	Load Balancer, Distributed Servers, IBM Cloud High Availability Setup
5	Performance	Optimization techniques to handle high traffic and improve response time (e.g., caching, CDN, request handling).	Redis Cache, CDN (Cloudflare / IBM CDN), API Gateway, Asynchronous Processing

